

Osteomyelitis

Aaron J. Tande

Gina A. Suh

Douglas R. Osmon

Elie F. Berbari

James M. Steckelberg

Short View Summary

Classification

- Osteomyelitis can develop as the result of contiguous spread from adjacent soft tissues and joints, hematogenous seeding, or direct inoculation of microorganisms into the bone because of trauma or surgery.
- The Cierny-Mader classification is based on the affected portion of the bone, the physiologic status of the host, and the local environment.
- The Lew and Waldvogel classification is based on the duration of illness (acute vs. chronic), the mechanism of infection (hematogenous vs. contiguous), and vascular insufficiency.

General Principles

- The diagnosis of osteomyelitis is suspected on clinical grounds. Confirmation usually entails a combination of radiologic, microbiologic, and pathologic tests.
- Cross-sectional imaging modalities, such as computed tomography or magnetic resonance imaging (MRI), are considered standard of care in diagnosing osteomyelitis.
- Repeated swab cultures of the same organism from draining wounds and sinus tracts may be of diagnostic benefit. However, recovered organisms (except for *Staphylococcus aureus*) may not reflect the true pathogen.
- Surgical culture, bone biopsy, or needle aspiration is best for organism identification.
- Antimicrobial therapy alone is typically insufficient in the treatment of chronic osteomyelitis because of inadequate penetration of nonviable bone. Therapy is therefore comprised of a combination of complete surgical débridement and prolonged antimicrobial therapy (see [Table 106.3](#) and [Table 106.5](#)), with débridement being the mainstay of therapy.

Osteomyelitis After a Contaminated Open Fracture

- Contaminated open fractures can lead to the development of osteomyelitis of the fractured bone, typically at the fracture site in 3% to 25% of cases.
- Staphylococci and aerobic gram-negative bacilli are the two most common groups of microorganisms in this setting.
- The hallmark of osteomyelitis after open fracture is nonunion of the fracture site or poor wound

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- Staphylococci and aerobic gram-negative bacilli are the two most common groups of microorganisms in this setting.
- The hallmark of osteomyelitis after open fracture is nonunion of the fracture site or poor wound healing after wound closure or soft tissue coverage.
- Management of open contaminated fractures typically entails early aggressive wound irrigation and débridement, administration of parenteral and local antimicrobials, fracture fixation, and soft tissue coverage of exposed bone.

Vertebral Osteomyelitis and Spondylodiskitis

- The origin is hematogenous in most native cases; it may also occur postoperatively or postprocedurally.
- *S. aureus*, coagulase-negative staphylococci, and streptococci are the most common microorganisms encountered in pyogenic vertebral osteomyelitis.
- MRI of the spine is highly accurate in the diagnosis of vertebral osteomyelitis.
- Image-guided aspiration and biopsy is highly specific but lacks sensitivity.
- In contrast to other categories of osteomyelitis, most cases can be treated with a 6-week course of

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Osteomyelitis in Patients With Diabetes Mellitus or Vascular Insufficiency

- Patients typically have osteomyelitis contiguous with an ulcer of the lower extremity.
- The sensitivity of exposed bone or a probe-to-bone test was 60%, and the specificity was 91%.
- This category of osteomyelitis is often polymicrobial.
- Therapy typically requires a combination of surgical débridement and broad-spectrum antimicrobials.

Acute Hematogenous Osteomyelitis

- Acute hematogenous osteomyelitis of long bones occurs mainly in prepubertal children, intravenous drug abusers, and patients with indwelling central catheters.
- The most common recovered microorganism is *S. aureus*.
- Acute hematogenous osteomyelitis of long bones in prepubertal children is typically treated with a 2- to 3-week course of antimicrobial therapy.

Classification

Despite advances in surgical and medical management of osteomyelitis, it is still considered one of the most difficult-to-treat infectious diseases. Progressive destruction of the bone and the formation of sequestra (dead bone) are characteristics of this disease. Osteomyelitis can develop as the result of contiguous spread from adjacent soft tissues and joints, hematogenous seeding, or direct inoculation of microorganisms into the bone because of trauma or surgery.¹ When established, microorganisms produce a local inflammatory reaction that promotes bone necrosis and the formation of sequestra. *Staphylococcus aureus*, the most common microorganism recovered in osteomyelitis, preferentially causes this disease by expressing high-affinity adhesins to components of bone matrix that express fibronectin, laminin, collagen, or sialoglycoprotein. There are two major osteomyelitis classification schemes. Cierny and Mader² classified osteomyelitis based on the affected portion of the bone, the physiologic status of the host, and the local environment (**Table 106.1**). This classification lends itself to the surgical and medical treatment and prognosis of osteomyelitis. Stage 1 osteomyelitis typically can be treated with antimicrobial therapy alone, whereas stage 3 disease most often is managed with aggressive surgical débridement, antimicrobial therapy, and delayed orthopedic reconstruction. Lew and Waldvogel³ classified osteomyelitis based on the duration of illness (acute vs. chronic), the mechanism of infection (hematogenous vs. contiguous), and the presence of vascular insufficiency. In contrast to the Cierny and Mader classification, the Waldvogel classification is an etiologic classification and does not implicate a specific therapeutic strategy.

Because of differences in pathophysiology, diagnosis, and management, long bone osteomyelitis, osteomyelitis resulting from open fractures, vertebral osteomyelitis, osteomyelitis in patients with diabetes mellitus and peripheral vascular

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Because of differences in pathophysiology, diagnosis, and management, long bone osteomyelitis, osteomyelitis resulting from open fractures, vertebral osteomyelitis, osteomyelitis in patients with diabetes mellitus and peripheral vascular insufficiency, acute hematogenous osteomyelitis, and SAPHO (synovitis, acne, plantar pustulosis, hyperostosis, and osteitis) syndrome are discussed separately. In addition, osteomyelitis in specific hosts (e.g., sickle cell disease), in unusual locations (e.g., clavicle), and secondary to unusual microorganisms are reviewed.

Lessons From Experimental Models

Because of the lack of well-designed prospective clinical trials to guide the management of patients with osteomyelitis, recommendations about this disease have been primarily derived from experimental Page 1354 animal models, expert opinion, and retrospective cohort studies. Experimental models have been developed mainly to study the pathogenesis and treatment of osteomyelitis and offer a more controlled approach to a heterogeneous disease. ⁴

TABLE 106.1

Staging System of Osteomyelitis

Anatomic Type

- Stage 1: Medullary osteomyelitis
- Stage 2: Superficial osteomyelitis
- Stage 3: Localized osteomyelitis
- Stage 4: Diffuse osteomyelitis

Physiologic Class

- A Host: Normal host
- B Host: Systemic compromise (Bs)
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Local compromise (Bi)
Systemic and local compromise (Bis)
C Host: Treatment worse than the disease
Systemic or Local Factors That Affect Immune Surveillance, Metabolism, and Local Vascularity

Staging System of Osteomyelitis

Systemic (Bs)	Local (Bi)
Malnutrition	Chronic lymphedema
Renal, hepatic failure	Major vessel compromise
Diabetes mellitus	Small vessel disease
Chronic hypoxia	Vasculitis
Immune disease	Venous stasis
Malignancy	Extensive scarring
Extremes of age	Radiation fibrosis
Immunosuppression	Neuropathy
	Tobacco abuse

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From Mader JT, Shirliff M, Calhoun JH. Staging and staging application in osteomyelitis. *Clin Infect Dis*. 1997;25:1303–1309.

Pathogenesis in Experimental Models

In experimental models, normal bone is highly resistant to infection. Osteomyelitis in this setting can be created only after inoculation of large inocula and the creation of bone necrosis, which can result from bone trauma or surgery or can be caused by the presence of foreign bodies.^{5, 6} When digested by osteoblasts, *S. aureus* can survive in a metabolically less active and phenotypically altered state for a prolonged period. This small colony variant phenotype leads to impaired antimicrobial activity, particularly with antimicrobials that interfere with cell wall synthesis, and reduced production of proinflammatory molecules.^{7, 8} These characteristics might explain, in part, the high relapse rate of osteomyelitis treated with antimicrobials and the long incubation period.^{5, 6} Muller and associates⁹ studied leukocyte mobilization and phagocytosis in a guinea pig model of posttraumatic osteomyelitis. In this model, leukocyte locomotion was reduced after trauma and infection with *S. aureus* for 90 days.⁹

Antimicrobial Therapy in Experimental Models

Norden and coworkers¹⁰ analyzed the effect of the duration of antimicrobial therapy on the rate of bone sterilization in an experimental model of chronic *S. aureus* osteomyelitis that did not include surgical débridement as part of the experimental treatment. Of cultures, 78% and 16% yielded bacterial growth after 14 and 28 days of clindamycin therapy, respectively. These data support the need for a prolonged course of antimicrobial therapy in osteomyelitis. To our knowledge, the optimal

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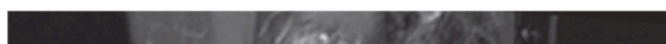
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Given the importance of biofilm production and the role of rifampin in killing microorganisms present in biofilms, several animal models of staphylococcal osteomyelitis have analyzed the efficacy of rifampin, alone or in combination with β -lactams, vancomycin, macrolides, and quinolones. Combinations of rifampin with other antimicrobial agents generally were more effective in sterilizing the bone. ¹¹ Rifabutin may be an alternative to rifampin if drug–drug interactions preclude the use of rifampin. ¹²

The effect of fluoroquinolones on the healing process of bone fractures was assessed by Huddleston and colleagues. ¹³ In this study, 60 rats were divided equally into three groups, each group receiving ciprofloxacin, cefazolin, or no treatment for 3 weeks. Radiographs revealed significantly impaired healing of fractures in the ciprofloxacin-treated group compared with fractures in the cefazolin-treated group. Additional studies have shown this effect with other fluoroquinolones but not with vancomycin or gentamicin. ^{14, 15} Quinolones should be used cautiously and when other antimicrobials are not available in cases of osteomyelitis in which fracture healing needs to occur concomitantly with antimicrobial administration. Given the inherent difficulties of conducting large prospective clinical trials in humans, the management of osteomyelitis is likely to continue to be guided by lessons learned from experimental animal models.



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FIG. 106.1 Magnetic resonance T2 axial image of the left tibia reveals abnormality of the middle and proximal portion of the tibia (white changes, arrows). During surgery, there was a significant amount of purulence and osteomyelitis bone present. Cultures grew *Pseudomonas aeruginosa*.



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[Follow for extended description](#)

General Principles of Diagnosis

Osteomyelitis can occur by hematogenous dissemination, contiguous spread from a soft tissue infection, or traumatic inoculation. Acute hematogenous infectious osteomyelitis in children is discussed separately. Adults usually have vague symptoms with a subacute-to-chronic presentation. Nonspecific pain around the involved site with the absence of systemic signs and symptoms is normal. Fever, chills, local swelling, and erythema in the proximity of the involved bone are seldom seen. A draining sinus tract may be present over the involved bone. It usually evolves over several months and sometimes years.

The diagnosis of osteomyelitis is first suspected on clinical grounds. Confirmation usually entails a combination of radiologic, microbiologic, and pathologic tests. The erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) can be elevated. The white blood cell count can be normal or elevated.

Although insensitive, a conventional radiograph is inexpensive and readily available. Abnormalities are usually seen 10 to 14 days after onset of the infection. Cross-sectional imaging modalities, such as computed tomography (CT) and magnetic resonance imaging (MRI), are considered standard of care in the diagnosis of osteomyelitis (Figs. 106.1 and 106.2).

Although expensive, they are very sensitive. In certain circumstances, such as Charcot arthropathy, these imaging modalities lack specificity. ¹⁶ ¹⁷ These modalities have an excellent anatomic delineation of the infectious process and an excellent resolution of the surrounding soft tissue. Given the high sensitivity of these tests, they are mostly used to rule out osteomyelitis. ¹⁷ Although the presence of orthopedic implants may limit the use of conventional MRI sequences, suppression techniques such as multiacquisition variable-resonance image combination (MAVRIC) improve image quality, allowing the use of MRI in this setting. ¹⁸ The presence of a cardiac device is not Page 1355 an absolute contraindication to MRI. Although patients with advanced renal disease may not be able to receive gadolinium contrast agents, MRI without contrast enhancement offers adequate diagnostic information. ¹⁹

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FIG. 106.2 Computed tomography reconstruction images of a nonunited femur fracture 6 months after intramedullary nailing. Surgical cultures grew *Pseudomonas aeruginosa* and *Cutibacterium acnes*.

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General Principles of Microbiologic Diagnosis

The identification of a causative microorganism is crucial to the optimal management of osteomyelitis. The type of organism and the associated in vitro susceptibility data help optimize medical therapy. Organism identification is best accomplished by means of surgical sampling or with needle aspiration under radiologic guidance to obtain tissues for histopathologic examination and aerobic and anaerobic bacterial culture. Swab cultures at the time of surgical sampling are not recommended. If a bacterial pathogen is not identified, less common causes of osteomyelitis should be considered and microbiologic evaluation should be guided by the clinical context and exposure history. Serologic testing and molecular diagnostics can be of benefit in cases caused by fastidious organisms or those that are culture negative (See “Culture-Negative Osteomyelitis”).

Swab cultures from draining wounds and sinus tracts can be of diagnostic benefit for three reasons. First, the identification of certain resistant microorganisms (e.g., methicillin-resistant *S. aureus* [MRSA], vancomycin-resistant enterococci) indicates the need for infection-control measures. Second, the isolation of *S. aureus* from superficial cultures has a high degree of correlation with deep cultures.²¹ However, the majority of sinus tract swabs obtained during *S. aureus* osteomyelitis will not grow this organism, highlighting the lack of sensitivity of this test. The recovery of other microorganisms correlates poorly with deep cultures. Third, the recovery of the same organism on two consecutive cultures, although infrequent, has been shown to highly correlate with deep surgical cultures of bones.²²

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TABLE 106.2

Microbiology of Osteomyelitis

Common (>50% of Cases)

Staphylococcus aureus

Coagulase-negative staphylococci

Occasionally Encountered (<25% of Cases)

Streptococci

Enterococci

Pseudomonas aeruginosa

Enterobacter spp.

Proteus spp.

Escherichia coli

Serratia spp.

Anaerobes (*Finegoldia* [*Peptostreptococcus*] spp., *Clostridium* spp., *Bacteroides fragilis* group)

Mycobacterium tuberculosis

Rarely Encountered (<5% of Cases)

Mycobacterium avium complex

Rapidly growing mycobacteria

Dimorphic fungi (blastomycosis, coccidioidomycosis, sporotrichosis)

Candida spp.

Cryptococcus

Aspergillus spp.

Mycoplasma spp.

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Aspergillus spp.

Mycoplasma spp.

Tropheryma whipplei

Brucella spp.

Salmonella spp.

Actinomyces

Bartonella spp.

Coxiella burnetii

Treponema pallidum

In hematogenous long bone osteomyelitis, the infection is usually monobacterial, whereas in contiguous infection it is frequently polymicrobial. *S. aureus*, coagulase-negative staphylococci, aerobic gram-negative bacteria, and *Finegoldia* (formerly *Peptostreptococcus*) spp. are the most common organisms encountered (**Table 106.2**). Community-acquired MRSA (CA-MRSA) has been increasingly recognized as a cause of acute long bone osteomyelitis and diskitis in children. ²³ Panton-Valentine leukocidin (PVL) is a cytotoxin found in CA-MRSA strains. In children, osteomyelitis caused by PVL-positive *S. aureus* is associated with multifocal and more aggressive disease when compared with PVL-negative *S. aureus*. ²⁴

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General Principles of Therapy

The goal of therapy for osteomyelitis is to eradicate the infection and to restore function. Most cases of osteomyelitis in adults require a combination of medical and surgical therapy for successful eradication of the infection. It has long been recognized that antimicrobial therapy alone is not curative in most cases of osteomyelitis. In 1944, with the introduction of penicillin, Key ²⁵ wrote: “continuous drug over a long period of time will lessen the amount of discharge, but it will not cure the disease because it cannot sterilize dead bone or cavities with necrotic content and rigid walls.” All antimicrobials should be withheld if possible until percutaneous aspirate or surgical deep cultures have been obtained, unless there is concomitant soft tissue infection or sepsis syndrome. Antimicrobials are usually started ([Table 106.3](#)) immediately after surgical débridement ([Table 106.4](#)).

Established Agents for Antimicrobial Therapy

β -Lactams and vancomycin are the most used antimicrobials in the medical management of osteomyelitis. ^{26, 27}

Cephalosporins are commonly used in patients with osteomyelitis because of their low toxicity profile and their spectrum of activity against staphylococci and other common bacterial pathogens that cause osteomyelitis. Cefazolin has excellent activity against methicillin-sensitive staphylococci, is safe and inexpensive, and has been used extensively in medical

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Salmonella spp.
Actinomyces
Bartonella spp.
Coxiella burnetii
Treponema pallidum

In hematogenous long bone osteomyelitis, the infection is usually monobacterial, whereas in contiguous infection it is frequently polymicrobial. *S. aureus*, coagulase-negative staphylococci, aerobic gram-negative bacteria, and *Finnegoldia* (formerly *Peptostreptococcus*) spp. are the most common organisms encountered (Table 106.2). Community-acquired MRSA (CA-MRSA) has been increasingly recognized as a cause of acute long bone osteomyelitis and diskitis in children. ²³ Panton-Valentine leukocidin (PVL) is a cytotoxin found in CA-MRSA strains. In children, osteomyelitis caused by PVL-positive *S. aureus* is associated with multifocal and more aggressive disease when compared with PVL-negative *S. aureus*. ²⁴

General Principles of Therapy

The goal of therapy for osteomyelitis is to eradicate the infection and to restore function. Most cases of osteomyelitis in adults require a combination of medical and surgical therapy for successful eradication of the infection. It has long been recognized that antimicrobial therapy alone is not curative in most cases of osteomyelitis. In 1944, with the introduction of penicillin, Key ²⁵ wrote: “continuous drug over a long period of time will lessen the amount of discharge, but it will not cure the disease because it cannot sterilize dead bone or cavities with necrotic content and rigid walls.” All antimicrobials should be withheld if possible until percutaneous aspirate or surgical deep cultures have been obtained, unless there is concomitant soft tissue infection or sepsis syndrome. Antimicrobials are usually started (Table 106.3) immediately after surgical débridement (Table 106.4).

Established Agents for Antimicrobial Therapy

β-Lactams and vancomycin are the most used antimicrobials in the medical management of osteomyelitis. ^{26, 27} Cephalosporins are commonly used in patients with osteomyelitis because of their low toxicity profile and their spectrum of activity against staphylococci and other common bacterial pathogens that cause osteomyelitis. Cefazolin has excellent activity against methicillin-sensitive staphylococci, is safe and inexpensive, and has been used extensively in medical therapy for osteomyelitis. ²⁶ The penicillinase-resistant penicillins have excellent antistaphylococcal activity and are commonly used, but carry a higher rate of adverse effects requiring alteration in therapy during outpatient parenteral antimicrobial therapy (OPAT), compared with cefazolin. ^{28, 29} Ceftriaxone once daily is convenient for outpatient therapy; its use in Page 1356 methicillin-sensitive staphylococcal osteomyelitis has been advocated by some experts. ^{26, 30}

TABLE 106.3

Antimicrobial Therapy for Chronic Osteomyelitis in Adults for Selected Microorganisms

MICROORGANISMS	FIRST CHOICE ^a	ALTERNATIVE CHOICE ^a
Staphylococci		
	Nafcillin sodium or oxacillin sodium. 1.5–2 g IV	Vancomycin. 15 mg/kg IV q12h for 4–

recognized that antimicrobial therapy alone is not curative in most cases of osteomyelitis. In 1944, with the introduction of penicillin, Key ²⁵ wrote: “continuous drug over a long period of time will lessen the amount of discharge, but it will not cure the disease because it cannot sterilize dead bone or cavities with necrotic content and rigid walls.” All antimicrobials should be withheld if possible until percutaneous aspirate or surgical deep cultures have been obtained, unless there is concomitant soft tissue infection or sepsis syndrome. Antimicrobials are usually started (**Table 106.3**) immediately after surgical débridement (**Table 106.4**).

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MICROORGANISMS	FIRST CHOICE ^a	ALTERNATIVE CHOICE ^a
Staphylococci		
Oxacillin sensitive	Nafcillin sodium or oxacillin sodium, 1.5–2 g IV q4–6h for 4–6 wk, <i>or</i> cefazolin, 2 g IV q8h for 4–6 wk, <i>or</i> ceftriaxone 2 g IV q24h for 4–6 wk ^b	Vancomycin, 15 mg/kg IV q12h for 4–6 wk; some add rifampin, 600–900 mg PO q day, to nafcillin/oxacillin
Oxacillin resistant (MRSA)	Vancomycin, 15 mg/kg IV q12h for 4–6 wk	Linezolid, 600 mg PO/IV q12h for 6 wk, <i>or</i> levofloxacin, ^d 750 mg PO/IV daily, plus rifampin, 600–900 mg/day PO for 6 wk if susceptible to both drugs
	<i>or</i> Daptomycin 6–10 mg/kg IV q24h ^c	
Penicillin-sensitive streptococci	Aqueous crystalline penicillin G, 20 × 10 ⁶ U/24 h IV either continuously or in 6 equally divided daily doses for 4–6 wk, <i>or</i> ceftriaxone, 2 g IV q24h for 4–6 wk, <i>or</i> cefazolin 2 g IV q8h for 4–6 wk	Vancomycin, 15 mg/kg IV q12h for 4–6 wk
Enterococci or streptococci with MIC ≥0.5 µg/mL	Aqueous crystalline penicillin G, 20 × 10 ⁶ U/24 h IV either continuously or in 6 equally divided daily doses for 4–6 wk, <i>or</i> ampicillin sodium, 12 g/24 h IV either continuously or in 6 equally	Vancomycin, 15 mg/kg IV q12h for 4–6 wk

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Enterococci or streptococci with MIC ≥0.5 µg/mL	Aqueous crystalline penicillin G, 20 × 10 ⁶ U/24 h IV either continuously or in 6 equally divided daily doses for 4–6 wk, <i>or</i> ampicillin sodium, 12 g/24 h IV either continuously or in 6 equally divided daily doses, <i>or</i> daptomycin 6–10 mg/kg IV q24h ^c	Vancomycin, 15 mg/kg IV q12h for 4–6 wk
Enterobacteriaceae	Ceftriaxone, 2 g IV q24h for 4–6 wk, <i>or</i> ertapenem 1 g IV q24h	Ciprofloxacin, ^d 500–750 mg PO q12h for 4–6 wk, <i>or</i> levofloxacin 500–750 mg PO q24h
<i>Pseudomonas aeruginosa</i>	Cefepime, 2 g IV q8–12h, meropenem, 1 g IV q8h <i>or</i> imipenem, 500 mg IV q6h <i>or</i> piperacillin-tazobactam 4.5 g IV q6h for 4–6 wk	Ciprofloxacin, ^d 750 mg PO q12h for 4–6 wk, <i>or</i> ceftazidime, 2 g IV q8h

MIC, Minimal inhibitory concentration; MRSA, methicillin-resistant *Staphylococcus aureus*.

^a Antimicrobial selection should be based on in vitro sensitivity data, allergies, and intolerances and on renal and hepatic function.

^b There is not universal consensus regarding use of ceftriaxone for staphylococcal osteomyelitis. Ceftriaxone should not be used with concomitant *Staphylococcus aureus* bloodstream infection.

^c Some experts recommend use of 8–10 mg/kg IV q24h.

Enterococci or streptococci with MIC ≥ 0.5 $\mu\text{g/mL}$	Aqueous crystalline penicillin G, 20×10^6 U/24 h IV either continuously or in 6 equally divided daily doses for 4–6 wk, or ampicillin sodium, 12 g/24 h IV either continuously or in 6 equally divided daily doses, or daptomycin 6–10 mg/kg IV q24h ^c	Vancomycin, 15 mg/kg IV q12h for 4–6 wk
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- ^d Should be avoided, if possible, in pediatric patients and in osteomyelitis associated with fractures.

TABLE 106.4

Surgical Principles in Osteomyelitis

Adequate drainage of all infected tissue
 Extensive debridement of all infected tissue
 Removal of all hardware if possible (see text)
 Management of dead space (flap)
 Complete wound closure
 Stability of infected fracture

Vancomycin is used commonly in the treatment of osteomyelitis because of its activity against methicillin-resistant staphylococci and ampicillin-resistant enterococci. In a large cohort study of 450 patients with osteomyelitis who were followed for 10 years, vancomycin was associated with a 2.5-fold relative risk of recurrence, compared with a penicillinase-resistant penicillin in a univariate analysis. ²⁶ Because of the high failure rate and the increasing minimal inhibitory concentration among many staphylococcus strains, many experts now advocate the use of higher and continuous dosage of vancomycin, keeping trough levels between 15 and 20 $\mu\text{g/mL}$. ^{31, 32} Further discussion of dosing based on trough levels and the use of the area under the curve (AUC) to minimum inhibitory concentration (MIC) ratio is discussed in **Chapter 29**. However, the data suggesting that higher trough levels improve outcomes are lacking, and rates of acute kidney injury are

function.

- b** There is not universal consensus regarding use of ceftriaxone for staphylococcal osteomyelitis. Ceftriaxone should not be used with concomitant *Staphylococcus aureus* bloodstream infection.
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Daptomycin, a cyclic lipopeptide antimicrobial agent, has bactericidal activity against aerobic and facultative gram-positive pathogens. The role of daptomycin in the therapy of patients with osteomyelitis is now established. ^{39, 40} A multicenter retrospective study in Europe of 220 patients with osteomyelitis evaluated the efficacy and safety of daptomycin. Only 0.5% of patients had an elevation of creatine phosphokinase, and 75% of patients had a successful outcome. ³⁹ However, resistance to daptomycin has occurred during prolonged therapy. ⁴¹ The role of daptomycin in osteomyelitis caused by methicillin-susceptible *S. aureus* osteomyelitis compared with β-lactam has not been assessed.

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Oral Agents for Antimicrobial Therapy

Although parenteral antimicrobial therapy has been the historical approach to treatment of osteomyelitis, ⁴² multiple recent studies provide evidence that oral antimicrobial therapy is equally effective. The OVIVA trial, a landmark pragmatic randomized controlled trial published in 2019, found no difference in outcomes between oral and parenteral therapy for bone and joint infections, including 415 patients with vertebral osteomyelitis or chronic osteomyelitis. ⁴³ Previous and subsequent studies, many of them observational, have supported the use of oral antimicrobial therapy for osteomyelitis. ⁴⁴⁻⁴⁶ This approach has the potential for significant cost savings and avoidance of adverse events that may accompany prolonged parenteral therapy. ^{44· 47· 48}

In the OVIVA trial and subsequent retrospective studies of oral therapy, the most commonly used oral agents included quinolones and rifampin (in combination therapy), although tetracyclines, macrolides/lincosamide, and penicillins were used less frequently. ⁴³⁻⁴⁵ First-generation quinolones, such as ciprofloxacin, have reduced activity against staphylococci, streptococci, and enterococci and no activity against anaerobes. Levofloxacin has good streptococcal and staphylococcal

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TABLE 106.5

Oral Antimicrobial Therapy for Chronic Osteomyelitis in Adults for Selected Microorganisms

MICROORGANISMS	FIRST CHOICE ^a	ALTERNATIVE CHOICE
Staphylococci ^b		
Oxacillin sensitive	Levofloxacin 750 mg PO daily, <i>or</i> cefadroxil 1000 mg PO bid	Minocycline/doxycycline 100 mg PO bid, <i>or</i> trimethoprim-sulfamethoxazole, ^d <i>or</i> clindamycin 300–450 mg PO q6h
	PLUS	PLUS
	Rifampin 600 mg PO bid	Rifampin 600 mg PO bid

subsequent studies, many of them observational, have supported the use of oral antimicrobial therapy for osteomyelitis.⁴⁴⁻⁴⁶ This approach has the potential for significant cost savings and avoidance of adverse events that may accompany prolonged parenteral therapy.^{44, 47, 48}

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	<i>PLUS</i>	<i>PLUS</i>
	Rifampin, 600–900 mg/day PO ^c	Rifampin, 600–900 mg/day PO ^c
Oxacillin resistant	Levofloxacin 750 mg PO daily, <i>or</i> minocycline/doxycycline 100 mg PO bid, <i>or</i> trimethoprim-sulfamethoxazole, ^d <i>or</i> clindamycin 600 mg PO tid	Linezolid 600 mg PO bid ^e
	<i>PLUS</i>	
	Rifampin, 600–900 mg/day PO ^c	
Penicillin-sensitive streptococci	Amoxicillin 1000 mg PO tid, <i>or</i> cefadroxil 1000 mg PO bid	
Penicillin-sensitive Enterococci	Amoxicillin 1000 mg PO tid, <i>or</i> linezolid 600 mg PO bid ^e	
Enterobacteriaceae	Levofloxacin 500–750 mg PO q24h, <i>or</i> ciprofloxacin, 500–750 mg PO q12h	Trimethoprim-sulfamethoxazole ^d
<i>Pseudomonas</i>	Levofloxacin 750 mg PO q24h, <i>or</i> ciprofloxacin	

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Penicillin-sensitive streptococci	Amoxicillin 1000 mg PO tid, <i>or</i> cefadroxil 1000 mg PO bid	
Penicillin-sensitive Enterococci	Amoxicillin 1000 mg PO tid, <i>or</i> linezolid 600 mg PO bid ^e	
Enterobacteriaceae	Levofloxacin 500–750 mg PO q24h, <i>or</i> ciprofloxacin, 500–750 mg PO q12h	Trimethoprim-sulfamethoxazole ^d
<i>Pseudomonas aeruginosa</i>	Levofloxacin 750 mg PO q24h, <i>or</i> ciprofloxacin 750 mg PO q12h	

^a Antimicrobial selection should be based on in vitro sensitivity data, allergies, intolerances, and renal and hepatic function.

^b Rifampin has been used for staphylococcal infection in most of the studies evaluating oral therapy, even in the absence of orthopedic implants. The most frequently used combination is levofloxacin and rifampin. In the case of rifampin resistance or intolerance, each of the listed agents (except levofloxacin) may be used as a single agent, although the data supporting this approach are less robust.

^c The optimal dosing approach to rifampin is not known. The authors utilize rifampin dosed twice daily. Drug interactions between rifampin and other medications, including other antibiotics, need to be identified and mitigated.

^d When used for treatment of osteomyelitis, the authors use trimethoprim-sulfamethoxazole 8–10 mg/kg of trimethoprim component per day divided into 2–3 doses per day. Periodic monitoring of complete blood count, creatinine and potassium should be performed.

^e Linezolid may not be tolerated beyond 6 weeks because of bone marrow suppression.

When applying the results of the OVIVA and other studies to treatment of osteomyelitis, the prominent role of quinolone therapy and use of rifampin combination must be recognized. Further robust data on the use of other oral agents will lend

Other Forms of Osteomyelitis: Specific Hosts, Unusual Locations, or Unusual Organisms

Pubic Osteomyelitis

Pubic osteomyelitis, infection of the pubic symphysis and adjacent pubic bones, has been reported after a variety of urologic and gynecologic surgical procedures, including Marshall-Marchetti-Krantz urethropexy, prolonged catheterization, inguinal hernia repair, vaginal delivery, and prostatectomy or radiotherapy for prostate cancer. ¹⁵⁸⁻¹⁶⁵ *S. aureus*, *Enterococcus* spp., *E. coli*, *Pseudomonas* spp., and *Proteus* spp. are the most encountered organisms. Pubic osteomyelitis has also been reported to occur because of presumed hematogenous seeding of an inflamed pubic symphysis joint in athletes with a history of frequent strenuous ¹⁶⁶ exercise.

The main differential diagnosis that is generally considered in patients with possible pubic osteomyelitis is a sterile form of pubic symphysis inflammation referred to as osteitis pubis. It is believed that osteitis pubis can be triggered by repeated strenuous exercise, bone infarction, or surgery. That stated, one report found that most individuals that develop inflammatory signs and symptoms of the pubic bone after urogynecologic surgery have bacterial infection when bone biopsies are performed. ¹⁶⁷

Most patients with pubic osteomyelitis present with suprapubic pain and difficulty and pain with ambulation. In one study, the time between surgery and the diagnosis ranged from 2 to 18 months. ¹⁶⁸ Fever and leukocytosis are often not present ^{167, 169} though an elevated ESR (>20 mm/h) is present in 67% of patients. ¹⁶⁷ Plain radiographs may be normal early in the disease. Radiographs performed 6 months later can reveal pubic bone sclerosis, widening of the joint spaces, and rarefaction. CT and MRI are more sensitive than plain radiography and can define soft tissue much better. Fine-needle aspiration is sometimes helpful and should be considered in suspected cases of pubic osteomyelitis in which operative cultures are not going to be obtained. In addition, fine-needle aspiration may help differentiate cases of sterile osteitis pubis, including those that do not improve with conservative management.

True aseptic osteitis pubis can be managed with rest, nonsteroidal antiinflammatory drugs, and corticosteroids. Antimicrobial therapy should be administered in all other cases. In one study, surgical débridement of infected bone was required in more than 70% of cases. ¹⁶⁷

Osteomyelitis of the Clavicle

Osteomyelitis of the clavicle represents less than 3% of osteomyelitis cases. ^{139-141, 170} It may be hematogenous or related to subclavian vein catheterization or neck surgery. ¹⁷¹ *S. aureus* is the most common organism. ¹²⁸ A variety of other microorganisms have been described, including gram-negative bacteria and *M. tuberculosis*. ^{170, 172} Given the nontraumatic nature, most cases of clavicular osteomyelitis present a therapeutic challenge. The disease may manifest with acute local pain and swelling with positive blood cultures (i.e., *S. aureus*) or may be more chronic and indolent in nature (i.e., *M. tuberculosis*). In one study, the clinical duration of the symptoms ranged from 2 weeks to 1.5 years. ¹⁷³ All patients typically presented with clavicular site pain. Fever, localized swelling, or a mass (as well as soft tissue abscesses) were present in 60%, 30%, and 30% of cases, respectively. ¹⁷³ Plain radiographs of the clavicle can show sclerotic or lytic changes. Acute cases secondary to *S. aureus* can be treated with parenteral antimicrobial therapy alone (see [Table 106.3](#)). Chronic cases should be treated with surgical débridement, followed by antimicrobial therapy.

Osteomyelitis in Hemodialysis Patients

Metastatic infections complicate 7% of tunneled catheter-related bloodstream infections in dialysis patients. These metastatic infections usually affect the disk space, epidural space, and joint space. ¹⁷⁴ Because of a higher rate of *S. aureus* colonization in this patient population, most cases of hematogenous osteomyelitis are caused by this microorganism.

Other Forms of Osteomyelitis: Specific Hosts, Unusual Locations, or Unusual Organisms

Pubic Osteomyelitis

Pubic osteomyelitis, infection of the pubic symphysis and adjacent pubic bones, has been reported after a variety of urologic and gynecologic surgical procedures, including Marshall-Marchetti-Krantz urethropexy, prolonged catheterization, inguinal hernia repair, vaginal delivery, and prostatectomy or radiotherapy for prostate cancer. ^{158- 165} *S. aureus*, *Enterococcus* spp., *E. coli*, *Pseudomonas* spp., and *Proteus* spp. are the most encountered organisms. Pubic osteomyelitis has also been reported to occur because of presumed hematogenous seeding of an inflamed pubic symphysis joint in athletes with a history of frequent strenuous ¹⁶⁶ exercise.

The main differential diagnosis that is generally considered in patients with possible pubic osteomyelitis is a sterile form of pubic symphysis inflammation referred to as osteitis pubis. It is believed that osteitis pubis can be triggered by repeated strenuous exercise, bone infarction, or surgery. That stated, one report found that most individuals that develop inflammatory signs and symptoms of the pubic bone after urogynecologic surgery have bacterial infection when bone biopsies are performed. ¹⁶⁷

Most patients with pubic osteomyelitis present with suprapubic pain and difficulty and pain with ambulation. In one study, the time between surgery and the diagnosis ranged from 2 to 18 months. ¹⁶⁸ Fever and leukocytosis are often not present ^{167, 169} though an elevated ESR (>20 mm/h) is present in 67% of patients. ¹⁶⁷ Plain radiographs may be normal early in the disease. Radiographs performed 6 months later can reveal pubic bone sclerosis, widening of the joint spaces, and rarefaction. CT and MRI are more sensitive than plain radiography and can define soft tissue much better. Fine-needle aspiration is sometimes helpful and should be considered in suspected cases of pubic osteomyelitis in which operative cultures are not going to be obtained. In addition, fine-needle aspiration may help differentiate cases of sterile osteitis pubis, including those that do not improve with conservative management.

True aseptic osteitis pubis can be managed with rest, nonsteroidal antiinflammatory drugs, and corticosteroids. Antimicrobial therapy should be administered in all other cases. In one study, surgical débridement of infected bone was required in more than 70% of cases. ¹⁶⁷

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Osteomyelitis in Patients With Sickle Cell Disease

Patients with sickle cell disease are at increased risk for osteoarticular infections. Sites of bony infarction result from recurrent episodes of vasoocclusion in an expanded marrow space. These sites of necrotic bone are at risk for hematogenous seeding. ¹⁷⁵ Acute and long bone osteomyelitis and septic arthritis are the most commonly encountered syndromes. In a large cohort study of 299 patients with homozygous sickle cell anemia in France, the prevalence of osteomyelitis was 12%. ¹⁷⁶ *Salmonella* spp. and *S. aureus* remain the most commonly encountered microorganisms in sickle cell anemia patients with osteomyelitis. ¹⁷⁷ Most patients are children. ^{177, 178} Although osteomyelitis could be multifocal in this setting, long bones are commonly affected.

The differentiation between bone infarction and osteomyelitis can be challenging because their clinical and radiologic presentations are similar. A history of focalized bone pain and prolonged leukocytosis suggests osteomyelitis. ¹⁷⁸ Sequential radionuclide bone marrow and bone scans can aid in the differentiation process. ¹⁷⁹ CT-guided aspirate or an open biopsy with cultures sometimes is needed for a more definitive diagnosis, but the procedure can infect previously sterile, infarcted bone. Surgical and medical therapies are similar to those for osteomyelitis in patients without sickle cell disease. Empirical antimicrobial therapy should be directed against *Salmonella* and *S. aureus*. ¹⁸⁰

Gaucher Disease

An acute bone crisis, typically affecting the tibia in patients with Gaucher disease, the most prevalent lysosomal storage disorder, is accompanied by fever, intense local pain, and signs of inflammation. Intravascular activation of coagulation is postulated to be the cause of the ischemic insult to the bone. Bone crisis may occur in stable and treated patients with enzyme replacement therapy. Like patients with SS-hemoglobin disease (sickle cell anemia), these patients may be incorrectly considered to have acute osteomyelitis and, if biopsy is performed, may be at risk for developing osteomyelitis at the biopsy site. ¹⁸¹

Osteomyelitis in Injection Drug Users

Osteoarticular infections occur more commonly in injection drug users (see [Chapter 316](#)). ¹⁸² Pathogens can reach the

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Osteomyelitis in Injection Drug Users

Osteoarticular infections occur more commonly in injection drug users (see [Chapter 316](#)).¹⁸² Pathogens can reach the bones via hematogenous routes or through contiguous or direct inoculation. Multiple skeletal sites can be affected. Unusual sites of infection outside this setting are common in these patients, such as sternoclavicular, sternochondral joint, sacroiliac joint, and pubic symphysis. *S. aureus*, *Pseudomonas* spp., and *Candida* spp. are the most encountered organisms in these patients. *M. tuberculosis* can cause vertebral osteomyelitis in these patients. *Eikenella corrodens*, a normal oral flora microorganism, can cause osteomyelitis in injection drug users who lick the needle tip or the skin before injection (needle licker osteomyelitis).¹⁸² Surgical and medical therapy for osteomyelitis in these patients is similar to therapy in other groups with osteomyelitis. Although OPAT and central venous access catheters should be used with caution in these patients, emerging literature suggests that OPAT may be similarly safe and effective for people who inject drugs.¹⁸³

Skeletal Mycobacterial Infection

Extrapulmonary disease represents 20% of all tuberculosis (see [Chapter 255](#)). Infection of the musculoskeletal system represents 1% to 5% of all tuberculosis cases. Osteomyelitis caused by *M. tuberculosis* often affects the spine or a paraarticular focus. Most cases are the result of a hematogenous spread from a pulmonary source. ^{184, 185}

The clinician should consider tuberculous osteomyelitis in patients with a past medical history of treated or untreated tuberculosis with new back pain, patients with a known positive tuberculin skin test or interferon gamma release assay (IGRA), young patients, patients coming from endemic areas with chest radiographic findings consistent with active tuberculosis or old healed tuberculosis, patients with a household member who had tuberculosis, patients with negative bacterial cultures, or patients whose biopsy specimen of infected bone shows granulomatous inflammation. Clinical features of osteomyelitis caused by *M. tuberculosis* are pain and swelling with abscess and sinus formation. Radiographs reveal irregular cavities and areas of bone destruction with little surrounding sclerosis. Because of the presence of a sinus tract, secondary bacterial infection does occur, although infrequently. ¹⁸⁵

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Vertebral Tuberculosis (Pott Disease)

Vertebral osteomyelitis caused by *M. tuberculosis*, also called *Pott disease*, is among the most common osteoarticular manifestations of tuberculosis. In this form of vertebral osteomyelitis, in contrast to bacterial vertebral osteomyelitis, systemic symptoms are often absent. Back pain or stiffness is commonly the only symptom, and a delay in the diagnosis is often the norm. In 50% of patients with spinal tuberculosis, MRI reveals paravertebral soft tissue abscesses in addition to the bone lesion. The infection has a predilection for the anterior superior or inferior angles of the vertebral bodies, especially in the early phases of the disease. ^{184, 186} Characteristics and challenges of the diagnosis and course of the disease are presented in a multinational report of 314 patients with a presumed diagnosis of spinal tuberculosis from high-prevalence countries (Egypt, Albania, Greece, and Turkey). ¹⁸⁷ Time from onset of symptoms to diagnosis was 78 days. Lumbar disease was most common (56%), followed by thoracic (49%) and thoracolumbar involvement (13%). Of 200 histopathologic examinations, 74% were consistent with tuberculosis, but a causative agent was cultured in only 41%. ¹⁸⁷ Mortality was 2%, but 25% developed sequelae, including structural abnormalities (kyphosis, gibbus deformity, scoliosis) and neurologic problems (paraparesis, paraplegia, and loss of sensation). ¹⁸⁷ In contrast, in Denmark, the time from symptoms to diagnosis of tuberculosis was 19.5 days for spinal disease and 28 days for other bone and joint manifestations. Culture was diagnostic in 87%. Sequelae were present in 57% of spinal cases, with overall mortality of 25.5% among Danes compared with 1.3% among immigrants. ¹⁸⁸

Diagnosis of Tuberculous Osteomyelitis

Significant overlap in imaging appearances between tuberculous osteomyelitis and other forms of osteomyelitis exists. The diagnosis should rely on the presence of *M. tuberculosis* confirmed by stain, culture, or molecular diagnostics of a biopsy specimen. Chest radiographs show an abnormality in less than 50% of patients with musculoskeletal tuberculosis, but should be obtained routinely because the existence of concomitant pulmonary tuberculosis has infection-control ramifications and may provide for an alternative area from which to obtain culture specimens. Therapy for skeletal tuberculosis is discussed in [Chapter 255](#).

Nontuberculous Mycobacterial Osteomyelitis

Osteoarticular infections with nontuberculous mycobacteria can also occur. They are commonly seen in immunocompromised patients ¹⁸⁹ or after contamination of a wound after trauma or surgery. *Mycobacterium marinum*, *M.*

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Medical therapy alone is often curative, although in selected cases, surgical débridement is required. Antimicrobial agents typically used in the treatment of osteoarticular infection caused by atypical mycobacteria are the same as agents used to treat infection at other sites and are discussed in [Chapters 257](#) and [258](#).

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Brodie Abscess

Brodie abscess refers to a chronic localized bone abscess. Patients with subacute cases may present with fever, pain, and periosteal elevation, whereas patients with chronic Brodie abscess are often afebrile and present with long-standing dull pain. The most common site of involvement is the distal part of the tibia. The lesion is typically single and located near the metaphysis. Of patients, 75% are younger than 25 years. Surgical débridement and culture-directed antibiotics are often curative. Cultures may be negative.

Culture-Negative Osteomyelitis

Rarely, bone culture specimens are sterile despite clinical, radiologic, and pathologic evidence of osteomyelitis. Brodie abscess and bone infarcts caused by Gaucher disease or sickle cell disease should be considered. In our practice, most of these cases are because of prior use of antimicrobial therapy. For indolent cases not responding to therapy, consideration should be given to stopping antibiotics and waiting for at least 2 to 4 weeks before repeating the culture. ²⁰⁶ Antimicrobial therapy may slow the growth of ordinarily hardy organisms. ²⁰⁷ When aerobic and anaerobic bacterial specimens are sterile, cultures should be obtained for fungi and mycobacteria. In selected cases, polymerase chain reaction analysis using 16S rRNA gene primers with a broad specificity for detecting bacterial DNA in bone and purulent material can be helpful. ^{207, 208} If all cultures are negative, we believe that the antimicrobial regimen should be designed to cover the commonly encountered organisms that are clinically suspected, taking into account the history of prior use of antimicrobial agents.

Summary

Despite important medical and surgical advances in management of patients, osteomyelitis remains extremely difficult to treat. The relapse rate can be as high as 20%. The optimal management of osteomyelitis requires a multidisciplinary health care team, including an orthopedic surgeon, neurosurgeon, oral surgeon, plastic surgeon, vascular surgeon, invasive radiologist, microbiologist, primary care provider, and infectious disease specialist. The usual goal of therapy is the eradication of the infection and restoration of function. Treatment of chronic osteomyelitis usually requires aggressive surgical débridement and appropriate antimicrobial therapy.

Acknowledgement

Drs. Tande, Suh, Osmon, and Berbari would like to acknowledge James Steckelberg (1954–2020), MD, for his many contributions to the management of osteomyelitis. An original member of the Mayo Clinic Orthopedic Infectious Diseases Focus Group and author of this chapter for many years, Dr. Steckelberg was instrumental to advancing the science of orthopedic infections. His clinical excellence improved the lives of many patients, and his dedicated mentorship will continue to influence future generations of physicians. Page 1365

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Orthopedic Implant–Associated Infections

Werner Zimmerli
Parham Sendi

Short View Summary

PERIPROSTHETIC JOINT INFECTION (PJI)

Diagnosis

- Determination that a prosthetic joint is infected depends on recovery of an organism from a joint aspirate or surgically obtained material from the joint. Synovial or tissue inflammation, wound dehiscence, and a draining sinus tract are also confirmatory signs of PJI. Joint effusion and early loosening of the prosthesis can point toward this diagnosis.

Microbiology

- The most frequent microorganisms are *Staphylococcus aureus*, coagulase-negative staphylococci, and streptococci.
- *Cutibacterium acnes* is the most important microorganism in periprosthetic shoulder infection.
- Microorganisms establish a biofilm on the surface of the implant.

Pathogenesis

- Foreign devices are covered by host proteins (e.g., fibronectin) after implantation, favoring bacterial adherence.
- The minimal infecting dose is greater than 10,000-fold lower in the presence than in the absence of an implant.
- Granulocyte function around the implant is impaired (activation and degranulation).

Clinical Manifestations

- Acute exogenous PJI produces local signs of inflammation (wound dehiscence, secretion, erythema).
- Acute hematogenous PJI involves new-onset pain at any time after implantation, initially without local signs of infection, and new-onset articular effusion.
- In chronic PJI, pain occurs because of early loosening, local inflammation, sinus tract, and chronic articular effusion.

Treatment

- The cornerstone of successful treatment is early diagnosis.
- Cure is possible only with adequate surgery combined with appropriate antibiotic therapy.
- A treatment algorithm (see [Fig. 107.1](#)) allows one to choose the most appropriate surgical intervention: débridement with retention, one-stage exchange, two-stage exchange, removal without replacement, or suppressive therapy.
- Guidelines are available at

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www.idsociety.org/Organ_System/#Skeletal%20%28Bones%20&%20Joints%29.

INTERNAL FIXATION–ASSOCIATED INFECTION

Diagnosis

- Diagnosis of infection associated with implanted devices used to stabilize bone fractures, such as pins and rods, requires culture of the device, usually after surgical removal. Infection related to fracture fixation is defined as fracture-related infection (FRI). The definition uses suggestive and confirmatory criteria, based on clinical, microbiologic, and histopathologic findings.

Microbiology

- Microbiology is comparable with that of PJI. Staphylococci are the most important pathogens.
- In open fractures, the wound is contaminated with environmental bacteria.

Pathogenesis

- The exogenous route plays a more important role than the hematogenous route.
- Infection may also occur via an adjacent focus (contiguous).

Clinical Manifestations

- Acute early postoperative infections cause wound healing disturbances, discharge, and erythema.
- In the case of acute symptoms that occur after an uneventful postoperative period, if systemic infection signs are dominating, pain is the most important local sign. Other local features are absent in the beginning and become apparent in the course of disease.
- Acute symptoms may also be caused by reactivation of chronic posttraumatic osteomyelitis that has been silent for many years. Pain may occur without prominent systemic inflammatory signs.
- Chronic symptoms include sinus tract, pain, and implant loosening.

Treatment

- Bone fractures are less susceptible to infection if stabilized.
- In acute infections, retain implant with débridement and antimicrobial therapy until the fracture is consolidated.
- In chronic infections, the implant must be removed. When delayed infections on implants are suppressed with antimicrobials, the hardware must be removed after healing of bone fracture.

Orthopedic devices are used to stabilize bone fractures, fuse the vertebral column, correct deformities such as scoliosis, and replace damaged joints. ¹⁻³ Whereas internal stabilization devices could be removed after a bone fracture is healed, prosthetic joints are kept in the body as long as their function remains full and pain free. The presence of an implant increases the risk for an infection because its local susceptibility for bacterial adherence is high. ⁴ Thus, even with the most meticulous infection control, implant-associated infections occur. The optimal management of orthopedic implant-associated infections requires close cooperation of experienced specialists in infectious diseases, microbiology, orthopedic and trauma surgery, and plastic surgery. ^{5,6}

Periprosthetic Joint Infection

Definition

The classic Centers for Disease Control and Prevention definition of wound infection is not ideal for implant-associated infections, ⁷ because superficial and deep wound infection cannot reliably be differentiated clinically. ⁸ Therefore, even in the case of a presumably “superficial” wound infection, revision surgery must be considered. Empirical antibiotic treatment without diagnostic workup should be avoided in the absence of life-threatening sepsis.

The definition of periprosthetic joint infection (PJI) should have a high sensitivity not to delay the diagnosis of PJI. For